

so that, by the assumption in (5.26b),

$$\limsup_n \left| \int_{t_0}^{t_n} u'_n(t) dt \right|^2 \leq \limsup_n \Gamma(\theta - 1) t_0^{1-\theta} E^\theta(u_n) = 0 ,$$

which contradicts (A.10) and concludes the assertion. $\square$

APPENDIX B. MEASURE-PRESERVING DIFFEOMORPHISMS

Let (M, g) be a smooth connected, orientable Riemannian manifold with Riemannian volume measure vol_g . We collect here some auxiliary results about the group $\text{Diff}_0^+(g)$ of all compactly non-identical, orientation-preserving, vol_g -preserving diffeomorphisms.

Firstly, let us recall that $\text{Diff}_0^+(g)$ is the (infinite-dimensional) Lie group corresponding to the Lie algebra of div_g -free vector fields on M with the Lie derivative as its Lie bracket. Let us further recall some virtually well-known results about the natural action of $\text{Diff}_0^+(g)$ on M .

The following may be easily inferred from the arguments in [13, §3].

Lemma B.1 (Extension lemma). *Let (M, g) be in addition open or boundaryless, and $K \subset M$ be any contractible compact subset. Then, every smooth vector field on K has a compactly supported, div_g -free extension to the whole of M .*

As an immediate consequence, we see that the Lie algebra of div_g -free vector fields is infinite-dimensional, thus so is $\text{Diff}_0^+(g)$.

Proposition B.2 (Transitivity, [13, Thm. A, §3, p. 98]). *$\text{Diff}_0^+(g)$ acts k -transitively on M for every $k \in \mathbb{N}_1$. In particular, it acts transitively on M .*

B.1. Actions on measures. In the following, let $B_1 \subset \mathbb{R}^d$ be the open unit ball equipped with the standard Euclidean metric g_e .

Lemma B.3. *Let ν be any finite measure on B_1 invariant for the $(\mathbb{Q}_+)$ -action of $\text{Diff}_0^+(g_e)$. Then $\nu \propto \mathcal{L}^d|_{B_1}$. In particular, ν has no singular continuous part.*

Proof. Let $Q_1, Q_2 \subset B_1$ be arbitrary closed cubes with equal volume, define K as the closed convex hull of $Q_1 \cup Q_2$, and let w be the vector field on K defining the translation of Q_1 to Q_2 . By Lemma B.1 applied to K and w , there exists a compactly supported div_{g_e} -free vector field w' on M extending w on K . Then, the flow of w' at time 1 is a compactly non-identical orientation-preserving, $\mathcal{L}^d$ -preserving diffeomorphism on M mapping Q_1 to Q_2 . In other words, $\text{Diff}_0^+(g_e)$ acts transitively on all closed cubes in B_1 , hence, by continuity, on the family $\mathcal{C}$ of all semi-closed cubes Q of the form $Q := x + q[0, 1)^d$ with $x \in \mathbb{R}^d$, $q \in \mathbb{Q}^+$ and such that $\overline{Q} \subset B_1$.

Since ν is invariant, by a standard decomposition argument, for every rational $q \in \mathbb{Q}^+$ and every $Q \in \mathcal{C}$ with $qQ \in \mathcal{C}$ we have $\nu(qQ) = q^d \nu Q$. Now, let $Q_0 \in \mathcal{C}$ be fixed, and set $c := \nu Q_0 / \mathcal{L}^d Q_0$. For every $Q_1 \in \mathcal{C}$ there exists $q \in \mathbb{Q}^+$ and $x \in \mathbb{R}^d$ so that $Q_1 = x + qQ_0$. For such q , by invariance of ν and of $\mathcal{L}^d$ (also under rescaling),

$$\nu Q_1 = \nu(qQ_0) = q^d \nu Q_0 = q^d c \mathcal{L}^d Q_0 = c \mathcal{L}^d(qQ_0) = c \mathcal{L}^d Q_1 .$$

Since $\mathcal{C}$ is a π -system generating the Borel σ -algebra of B_1 , it follows by a standard monotone class argument that $\nu = c \mathcal{L}^d$ as Borel measures on B_1 , which concludes the proof. $\square$